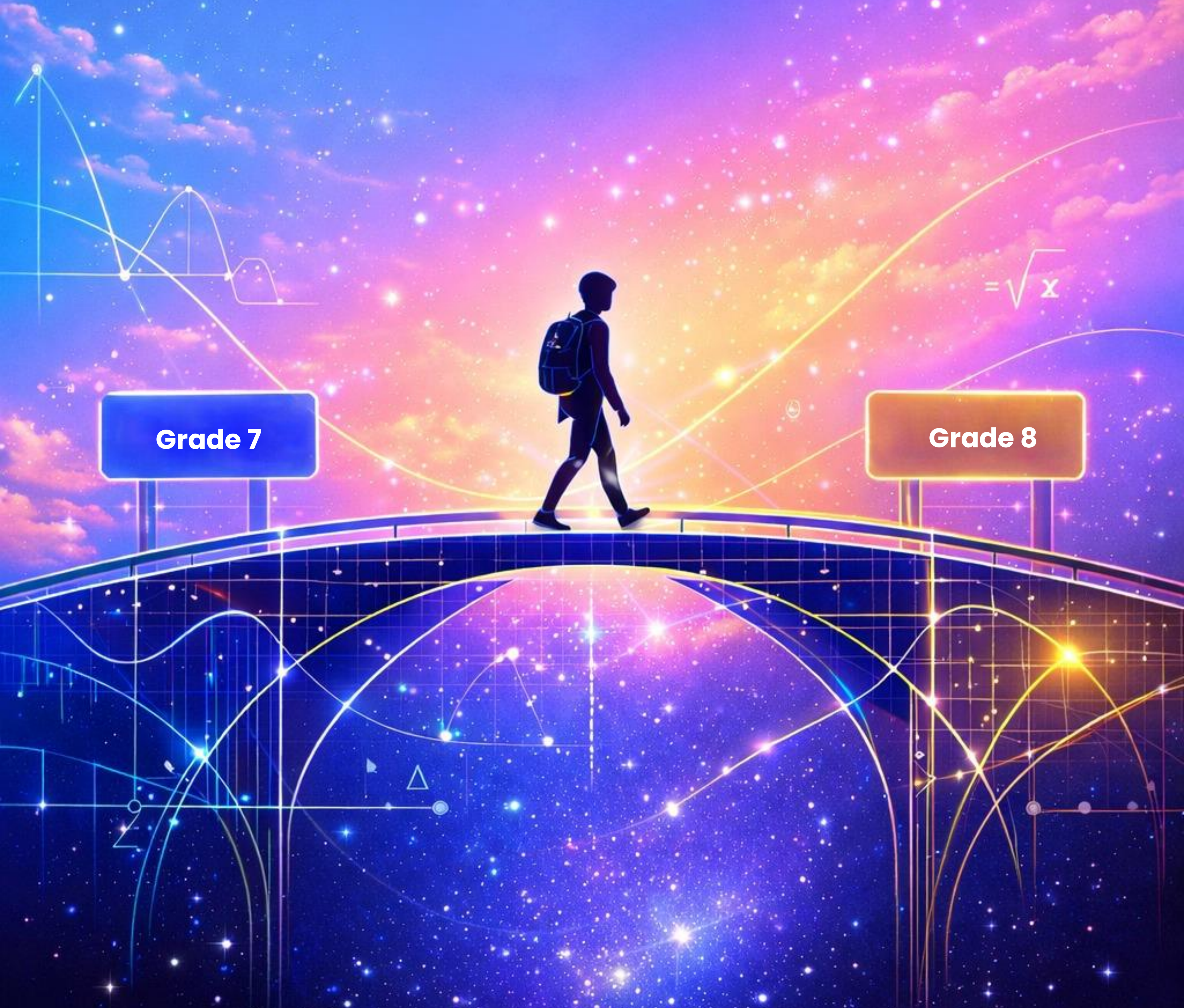
BRIDGE COURSE
GRADE – 8

CHEMISTRY



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1. NATURE OF MATTER AND ITS PHYSICAL STATES

1.1 Matter

Matter is anything that has mass and occupies space (volume).

For example: book, pen, water, air, clothes.

So, we can understand matter in a simple way as:

Matter = Mass + Volume

This means if something has mass and takes up space, it is matter. Matter exists everywhere around us and mainly occurs in three states — solids, liquids, and gases — depending on how its particles are arranged and move.

Difference between Mass and volume

Mass	Volume
Tells how much matter an object contains	The space taken up by matter.
Measured using a balance (units: g, kg).	Measured in mL, L (SI unit: m^3).

Important points:

- If something occupies space and can be measured for mass, it is matter.
- Solids, liquids, gases — all are matter because all have mass + volume.
- Many things that we can't see clearly (like air) are still matter because they occupy space and have mass.

1.2 Three States of Matter

Matter exists mainly in three states: **Solid, Liquid, and Gas.** These states differ in their shape, volume, and ability to flow.

(i) Solid

- **Shape:** Fixed (definite shape)
- **Volume:** Fixed (definite volume)
- **Flow:** Do not flow
- **General Properties:** Rigid and have a strong structure
- **Examples:** Stone, book, iron nail, chalk

Explanation:

In solids, particles are tightly packed and held together by strong forces. Therefore, solids maintain their shape and volume.

(ii) Liquid

- **Shape:** Not fixed — takes the shape of the container
- **Volume:** Fixed (definite volume)
- **Flow:** Flow easily
- **General Properties:** Particles are less tightly packed than solids
- **Examples:** Water, milk, oil

Explanation:

In liquids, particles can move around each other. That is why liquids can flow and take the shape of their container, but their volume remains the same.

(iii) Gas

- **Shape:** Not fixed
- **Volume:** Not fixed — fills the entire container
- **Flow:** Flow very easily
- **General Properties:** Particles are very far apart and move freely
- **Examples:** Air, oxygen, carbon dioxide

Explanation:

In gases, particles are far apart and move freely in all directions. Therefore, gases have neither fixed shape nor fixed volume.

Examples

(i) Which of the following is NOT considered matter?
(A) Air inside a balloon **(B) Steam (water vapour)**
(C) Heat **(D) Oxygen gas**

Sol. **(C) Heat**
Matter must have mass and occupy space.
Air, steam, oxygen → occupy space + have mass
Heat is a form of energy, it does not have mass/volume like matter

(ii) A substance has fixed volume but no fixed shape. It is most likely a:
(A) Solid **(B) Liquid**
(C) Gas **(D) Plasma**

Sol. **(B) Liquid**
Solid: fixed shape + fixed volume
Liquid: no fixed shape but fixed volume
Gas: no fixed shape + no fixed volume

(iii) Which statement is correct about gases?
(A) Gases have definite shape and definite volume
(B) Gases have definite shape but no definite volume
(C) Gases have no definite shape but definite volume
(D) Gases have no definite shape and no definite volume

Sol. **(D)** Gas particles spread out and fill the entire container, so both shape and volume are not fixed.

(iv) Using the terms mass and space, explain why air is matter, even though it is invisible.

Sol. Air is matter because it occupies space (air in an inverted glass prevents water from entering) and has mass (an inflated balloon is heavier than an empty one).

(v) Write two differences between solid, liquid and gas based on:
(i) Shape and **(ii) Compressibility.**
Give one example of each state.

Sol. **(i)** Shape
Solid: Fixed shape (example: stone/eraser)
Liquid: No fixed shape, takes shape of container (example: water/oil)
Gas: No fixed shape, spreads everywhere (example: air/oxygen)

(ii) Compressibility
Solid: Almost not compressible
Liquid: Slightly compressible (very low)
Gas: Highly compressible (can be squeezed easily)

Practice Questions - 1

(i) Which pair correctly matches the statement with matter?
(A) Occupies space only
(B) Has mass only
(C) Has mass and occupies space
(D) Has energy and occupies space

(ii) A gas kept in a closed container is transferred to a bigger container. What will change?
(A) Only shape
(B) Only volume
(C) Both shape and volume
(D) Neither shape nor volume

(iii) Which of the following shows that a liquid has fixed volume?
(A) It spreads in all directions
(B) It can be poured from one container to another without changing quantity
(C) It cannot flow
(D) It has a fixed shape

(iv) Which of the following is most compressible?
(A) Chalk piece
(B) Water in a bottle
(C) Air in a syringe
(D) Ice cube

(v) A student says: "All matter is visible." Which example proves this statement is wrong?
(A) Milk (B) Wood
(C) Air (D) Brick

(vi) Choose the correct order of ability to flow (from least to most):
(A) Solid < Liquid < Gas (B) Liquid < Solid < Gas
(C) Gas < Liquid < Solid (D) Solid < Gas < Liquid

(vii) Define matter in one sentence and write two examples to support your definition.
(Answer in 3–4 lines.)

(viii) A student says: "Gases do not occupy space because we cannot see them."
Write a short explanation (4–5 lines) to correct the student using an everyday example.

(ix) Write any three differences between liquid and gas based on:
(i) volume (ii) compressibility

(x) Give one example each of: solid, liquid, gas from your classroom or home.
Then write one property for each example that proves its state.

2. INTERCONVERSION OF STATES OF MATTER

The substance can exist in different states — solid, liquid, and gas — and can change from one state to another. This process is called the **interconversion of states of matter**.

"Same substance, different states — state changes happen mainly due to heating and cooling."

Heating generally changes:

Solid → Liquid → Gas

Cooling generally changes:

Gas → Liquid → Solid

Example: Ice changes into water when kept in a cup at room temperature. This proves that ice and water are the same substance in different states.

2.1 Melting (Solid → Liquid)

Melting is the process in which a solid changes into a liquid on heating.

Everyday Example: When ice is taken out of the freezer and kept outside, it melts and becomes water.

Key Points:

- Melting occurs when a solid absorbs heat from its surroundings.
- Only the state changes; the substance remains the same.
- Ice and water are both the same substance (water), but in different states.
- Melting is a reversible process.

Other Example: Candle wax melts when heated.

2.2 Evaporation (Liquid → Gas)

Evaporation is the process in which a liquid changes into its vapour state at any temperature below its boiling point.

Key Observation: Water kept on a steel plate disappears even though it cannot seep through steel.

Conclusion: Water changes into water vapour (gaseous state).

Examples:

- Drying of wet clothes
- Drying of a mopped floor
- Sweat drying from our skin
- Water sprinkled on a hot pan disappears quickly

Key Points:

- Evaporation happens continuously at room temperature.
- It occurs from the surface of the liquid.
- Water vapour is invisible.
- Steam appears visible because tiny water droplets form in it.

2.3 Condensation (Gas → Liquid)

Condensation is the process in which a gas changes into a liquid on cooling.

Key Observation: When a glass containing cold water and ice is kept outside, water droplets appear on its outer surface.

Explanation: Water vapour present in the air cools when it touches the cold surface and changes into liquid droplets.

Important Points:

- Droplets form due to condensation of water vapour from air.
- Water is not leaking from the glass.
- Condensation happens when gas loses heat.

Examples:

- Water droplets on a cold glass
- Dew drops on plants in the morning

2.4 Freezing (Liquid → Solid)

Freezing is the process in which a liquid changes into a solid on cooling.

Examples:

- Water kept in a freezer changes into ice
- Coconut oil solidifies in winter

Key Points:

- Freezing occurs when heat is removed from the liquid.
- Only the state changes; the substance remains the same.
- The process is reversible (ice can melt back into water).

Examples

(i) The change of state from solid to liquid is called:

(A) Freezing **(B) Condensation**
(C) Melting **(D) Evaporation**

Sol. (C) Melting
Melting = solid → liquid (example: ice → water).

(ii) Wet clothes dry mainly due to:

(A) Condensation **(B) Freezing**
(C) Evaporation **(D) Melting**

Sol. (C) Evaporation
In evaporation, liquid water changes into water vapour and leaves the cloth, so clothes dry.

(iii) Water droplets appear on the outer surface of a cold glass kept in open air because of:

(A) Melting of ice
(B) Evaporation of water from the glass
(C) Condensation of water vapour present in air
(D) Freezing of water vapour

Sol. (C) Condensation of water vapour present in air
The cold surface cools nearby water vapour in air, which changes into liquid droplets → condensation.

(iv) **A student says: "Water disappears from a steel plate because it seeps into the steel." Explain in 4–5 lines what actually happens, using the correct process name.**

Sol. Steel is not porous, so water cannot seep into it. Water disappears because it changes into water vapour. This change of liquid water into vapour is called evaporation. The vapour mixes with air and becomes difficult to see.

(v) **Write the correct term for each change:**
(i) Ice → Water
(ii) Water → Water vapour (on heating)
(iii) Water vapour → Water droplets (on cooling)
(iv) Water → Ice

Sol. (i) Melting
(ii) Boiling / Evaporation (heating causes liquid → gas)
(iii) Condensation
(iv) Freezing

Practice Questions - 2

(i) When water is sprinkled on a hot pan and it disappears quickly, the main process is:
(A) Condensation (B) Evaporation
(C) Freezing (D) Melting

(ii) Which change happens when water is kept in a freezer for a few hours?
(A) Condensation (B) Evaporation
(C) Freezing (D) Melting

(iii) Water vapour changing into tiny water droplets on cooling is called:
(A) Evaporation (B) Condensation
(C) Melting (D) Boiling

(iv) Which of the following is the correct sequence on heating ice?
(A) Ice → Steam → Water (B) Ice → Water → Steam
(C) Water → Ice → Steam (D) Steam → Water → Ice

(v) During which process does gas change to liquid?
(A) Condensation (B) Melting
(C) Freezing (D) Evaporation

(vi) Which statement is correct?
(A) Freezing is solid → liquid (B) Melting is liquid → solid
(C) Condensation is gas → liquid (D) Evaporation is gas → liquid

(vii) Write the meaning of interconversion of states in 3–4 lines using water as an example.

(viii) Explain (in points) what happens when you take ice out of a freezer and keep it on a table. Name the two changes involved.

(ix) Give one daily-life example each of:
(i) Melting
(ii) Freezing
(iii) Condensation
(iv) Evaporation
(Write in one line each.)

(x) A cold bottle taken out from the fridge becomes wet on the outside after some time. Name the process and explain from where the water comes (4–5 lines).

3. ELEMENTS, COMPOUNDS & MIXTURES

A pure substance is made up of only one kind of particles and has a fixed composition and definite properties. Pure substances are uniform throughout and cannot be separated into simpler substances by physical methods.

Pure substances are mainly of two types:

- Elements
- Compounds

3.1 Elements

An element is a pure substance made up of only one type of atom.

Key Points:

- It consists of only one kind of particles.
- It cannot be broken down into simpler substances by physical methods.
- Each element is represented by a symbol (for example: Fe, Cu, O).
- Elements are the basic building blocks of matter.

Examples: Iron, copper, aluminium, oxygen, carbon

3.2 Compounds

A compound is a pure substance formed when two or more elements chemically combine in a fixed ratio.

Key Points:

- Composition is always fixed.
- Properties of a compound are different from the elements forming it.
- Components cannot be separated by physical methods.
- Separation usually requires a chemical change.
- Compounds are uniform throughout.

Examples: Water, carbon dioxide, common salt

3.3 Mixtures (Impure Substances)

A mixture is formed when two or more substances are mixed together physically. Most of the matter around us exists as mixtures.

Examples: Air, soil, sea water, milk

Example from daily life: Grains, stones, and husk are mixed together and can be separated using handpicking or winnowing.

Characteristics of Mixtures

- A mixture contains two or more substances.
- Composition is variable (amount of each substance can change).
- Each component retains its own properties.
- Components can be separated by physical methods.

3.4 Common Separation Methods

- Handpicking
- Sieving
- Filtration

- Evaporation
- Magnetic separation
- Sedimentation and decantation

Sometimes, more than one method is needed to separate mixture components.

Why are mixtures called impure substances?

In science, “impure” means that more than one substance is present together. It does not mean dirty. Therefore, mixtures are called impure substances.

Examples

(i) Which of the following is a mixture?

(A) Distilled water **(B) Oxygen**

(C) Air **(D) Common salt**

Sol. **(C) Air**

Air contains multiple gases mixed together, so it is not a single form of matter.

(ii) A pure substance is best described as:

(A) Any substance that looks clean

(B) A substance made of only one kind of particles

(C) Any substance that can be separated by filtration

(D) Any substance found in nature

Sol. **(B) A substance made of only one kind of particles**

Pure substance = single type of particles / single form of matter.

(iii) Which method is most suitable to obtain salt from salt-water?

(A) Filtration **(B) Handpicking**

(C) Evaporation **(D) Sieving**

Sol. **(C) Evaporation**

Evaporation converts liquid to vapour and leaves dissolved solid behind.

(iv) Define

(i) pure substance **(ii) mixture.**

Give one example of each.

Sol. Pure substance: Single type of particles; single form of matter; fixed nature.

Example: element (iron/oxygen) or compound (water/salt).

(v) Why are mixtures called “impure substances” in science? Explain with one example.

Sol. In science, “pure” means only one substance/one kind of particles.

If more than one substance is present together, it becomes a mixture, hence called impure (not necessarily dirty).

Example: grains + stones is a mixture and can be separated by handpicking.

Practice Question - 3

(i) Which of the following is a pure substance?
(A) Soil (B) Air
(C) Copper (D) Lemonade

(ii) A compound is formed when:
(A) Two substances mix in any ratio
(B) Two or more elements combine chemically in a fixed ratio
(C) Two liquids are mixed and stirred
(D) A solid dissolves in a liquid

(iii) Which of the following is an element?
(A) Carbon dioxide (B) Water
(C) Iron (D) Common salt

(iv) Which statement is correct for a mixture?
(A) It has fixed composition
(B) It cannot be separated by physical methods
(C) It has variable composition
(D) It is always made of only one substance

(v) Which method is most suitable to separate stones from rice grains?
(A) Evaporation (B) Handpicking
(C) Filtration (D) Condensation

(vi) "Salt solution" (salt + water) is best classified as:
(A) Element (B) Compound
(C) Mixture (D) Pure substance

(vii) Which pair is correctly matched?
(A) Element—Water (B) Compound—Oxygen
(C) Mixture—Air (D) Mixture—Copper

(viii) Define element and compound in 2–3 lines each. Give one example of each.

(ix) Write three differences between a pure substance and a mixture (point-wise).

(x) Is milk a pure substance?
Give your answer with reason in 4–5 lines. Also mention whether it is a mixture or pure substance.

4. THE ELEMENTS: METALS & NON-METALS

4.1 Element

An element is a pure substance made of only one kind of atom and cannot be broken down into simpler substances by physical or chemical methods.

Key Points:

- Elements are the basic building blocks of matter.
- There are 118 known elements.
- Some elements occur naturally, while others are made artificially in laboratories.
- Each element has its own unique properties.

Examples: Iron, copper, oxygen, carbon

Why do we classify elements?

Since there are many elements with different properties, classification helps us to:

- Study elements in an organized way
- Compare similar elements
- Understand their properties easily
- Predict their behaviour and use

4.2 Main Classification of Elements

Elements are mainly classified into three groups:

(a) Metals

(b) (b) non-metals

(c) (c) Metalloid

(a) Metals: Metals are elements that usually show properties like hardness, lustre, and good conductivity.

General Characteristics of Metals:

- Usually shiny (lustrous)
- Hard and strong
- Malleable (can be beaten into sheets)
- Ductile (can be drawn into wires)
- Good conductors of heat and electricity

Examples: Iron, copper, aluminium, zinc, gold

Uses: Utensils, wires, tools, machinery

(b) Non-metals: Non-metals are elements that generally do not show the properties of metals.

General Characteristics of Non-metals:

- Usually dull (not shiny)
- Not malleable or ductile
- Poor conductors of heat and electricity
- Many are gases at room temperature

Examples: Oxygen, nitrogen, carbon, sulfur, chlorine

Uses: Breathing (oxygen), fertilizers (nitrogen), fuels (carbon)

(c) Metalloids: Metalloids are elements that show properties of both metals and non-metals.
General Characteristics of Metalloids:

- Show mixed properties
- May appear like metals but behave like non-metals in some situations

Examples: Silicon, boron, germanium, arseni

4.3 Occurrence of Metals

Do metals occur freely in nature?

Metals occur in nature in two forms:

(a) Free State (Native State)

Some metals are less reactive and can exist freely in nature.

Reason: They do not react easily with air or water.

Examples: Gold, silver, platinum, copper

(b) Combined State

Most metals are reactive and are found combined with other elements in the form of compounds.

Examples: Sodium, potassium, magnesium, calcium

These compounds are found in the Earth's crust.

4.4 Minerals

Minerals are naturally occurring substances in the Earth's crust that contain metals in combined form.

Metals in minerals may be present as:

- Oxides
- Sulphides
- Carbonates
- Chlorides

Examples of Minerals:

- Bauxite → Aluminium
- Haematite → Iron
- Zinc blende → Zinc
- Copper pyrite → Copper

4.5 Ores

An ore is a mineral from which a metal can be extracted easily and economically.

Important points:
All ores are minerals, but not all minerals are ores.

Difference Between Mineral and Ore

Mineral	Ore
Contains metal	Contains metal
Extraction may not be easy or economical	Metal can be extracted easily and economically
May or may not be useful for extraction	Always useful for extraction

Why is the concept of ore important?

Some minerals contain very little metal or have too many impurities, making extraction difficult or expensive. Therefore, only suitable minerals are chosen as ores for metal extraction.

Impurities in Ores (Gangue)

Ores contain unwanted materials such as:

- Sand
- Clay
- Rocks

These unwanted substances are called **gangue**.

Gangue is removed before extracting the metal using physical methods.

Examples

(i) Elements are mainly classified into:

(A) Minerals, ores, gangue **(B) Metals, non-metals, metalloids**

(C) Compounds, mixtures, solutions **(D) Solids, liquids, gases**

Sol. **(B)**

Classification of elements is commonly done as metals, non-metals, and metalloids.

(ii) Which of the following is correctly matched?

(A) Gold — ore **(B) Haematite — ore of iron**

(C) Oxygen — metal **(D) Copper — non-metal**

Sol. (B)

Haematite is an iron ore (a mineral from which iron can be extracted economically).

(iii) Which statement is true?

(A) All minerals are ores **(B) All ores are minerals**

(C) Ores do not contain metals **(D) Metals are always found in free state**

Sol. **(B)**

Ore is a type of mineral from which metal can be extracted conveniently. Hence all ores are minerals, but all minerals are not ores.

(iv) Differentiate between mineral and ore.

Sol. **Mineral**: Naturally occurring substance in Earth's crust containing a metal (in combined form).

Ore: A mineral from which the metal can be extracted easily and economically.

Key relation: All ores are minerals, but all minerals are not ores.

(v) Explain the two ways metals occur in nature, with one example each.

Sol. **Free (native) state:** Less reactive metals may occur in free form.

Example: Gold (also silver, platinum).

Combined state (as minerals): Most metals occur as compounds in minerals.

Example: Iron occurs in minerals like haematite.

Practice Exercise - 4

(i) A student claims that "Sulphur should be a metal because it is an element." Which statement best corrects the student?
(A) Sulphur is a metal because it forms ores
(B) Sulphur is a non-metal and shows non-metallic properties
(C) Sulphur is a metalloid because it is solid
(D) Sulphur is neither metal nor non-metal

(ii) In a mining area, two minerals are found. Mineral X contains a metal, but extracting it is very costly. Mineral Y contains the same metal and extraction is easy and economical. Which conclusion is correct?
(A) Both X and Y are ores (B) Only X is an ore
(C) Only Y is an ore (D) Neither X nor Y is a mineral

(iii) You want to place an element in the "metalloid" category. Which of the following would be the best example?
(A) Oxygen (B) Silicon
(C) Copper (D) Sulphur

(iv) Gold is sometimes found in the Earth's crust in the native (free) state. The most suitable reason is that gold:
(A) reacts very quickly with air and water
(B) is highly reactive and forms many compounds easily
(C) is less reactive, so it does not combine easily
(D) exists naturally only as gas

(v) A teacher asks students to identify a set that contains only metals, suitable for making wires/utensils. Which option is correct?
(A) Iron, copper, aluminium
(B) Oxygen, nitrogen, sulphur
(C) Silicon, boron, arsenic
(D) Carbon, chlorine, iodine

(vi) Which statement best describes a mineral in the context of occurrence of metals?
(A) A mineral is always a pure metal
(B) A mineral may contain a metal in the combined state in nature
(C) A mineral is always a liquid
(D) A mineral cannot contain any metal

(vii) A student is preparing a chart of "Metal → Ore." Which correct matching should be written?
(A) Bauxite — ore of iron
(B) Haematite — ore of aluminium
(C) Bauxite — ore of aluminium
(D) Zinc blende — ore of copper

(viii) Classify the following as metal / non-metal / metalloid:
Iron, Sulphur, Silicon, Copper, Oxygen.

(ix) Write a short note (4–5 lines):
"Why most metals are found in combined state in nature?"

(x) Write the statement and explain in 3–4 lines:
"All ores are minerals, but all minerals are not ores."

5. METHODS OF SEPARATION

Why do we separate substances?

In our daily life, many substances exist as mixtures. These mixtures may contain unwanted components or sometimes two useful substances mixed together. For example, muddy water contains mud as an impurity, and tea contains tea leaves that we do not want to drink. Therefore, we use different separation methods to remove unwanted substances or to obtain useful components separately.

5.1 Sedimentation and Decantation

(i) Sedimentation: Sedimentation is the process in which heavier, insoluble solid particles settle down at the bottom of a liquid when the mixture is left undisturbed for some time.

Conditions required for sedimentation:

- The solid must be insoluble in the liquid.
- The particles must be heavy enough to settle down.
- The mixture must be left undisturbed for sufficient time.

Example: When muddy water is left undisturbed, mud settles at the bottom.

(ii) Decantation: Decantation is the process of carefully pouring the clear liquid from the top into another container without disturbing the settled solid.

Example: When tea is left undisturbed, tea leaves settle at the bottom. The clear tea can be poured into another cup.

Uses of Decantation:

- Washing rice and pulses to remove dust
- Separating sand or mud from water
- Removing settled impurities from stored water

Limitation of Decantation:

- Fine particles may remain in the liquid.
- Very light particles may not settle easily.
- Complete separation may not always be possible.

5.2 Filtration

Filtration is the process of separating an insoluble solid from a liquid using a filter such as cloth, strainer, or filter paper.

Example: Tea leaves are separated from tea using a strainer.

How filtration works

- The filter has very small pores.
- Liquid passes through the pores.
- Solid particles are trapped on the filter.

If particles are very fine, filter paper is used because it has smaller pores.

Important Terms in Filtration:

Residue: The solid substance that remains on the filter paper.

Filtrate: The clear liquid that passes through the filter.

Difference Between Decantation and Filtration

Decantation	Filtration
Depends on settling of particles	Depends on filter pores
Less effective for fine particles	More effective separation
Some impurities may remain	Gives clearer liquid

5.3 Magnetic Separation

Magnetic separation is the process of separating magnetic substances from non-magnetic substances using a magnet.
Magnetic substances: Substances attracted by a magnet (example: iron)
Example: If iron nails are mixed with sawdust, a magnet can be used to attract and separate the iron nails.
Uses of Magnetic Separation:

- Separating iron pins or nails from sand or wood dust
- Separating iron scrap from waste in industries
- Used in recycling processes

Examples

(i) Muddy water is kept undisturbed in a bucket. After some time, mud settles at the bottom. This process is called:
(A) Filtration **(B) Sedimentation**
(C) Decantation **(D) Magnetic separation**

Sol. **(B) Sedimentation**
In sedimentation, heavier insoluble particles settle at the bottom when the mixture is left undisturbed.

(ii) Which method is most suitable to separate tea leaves from tea?
(A) Decantation **(B) Filtration**
(C) Magnetic separation **(D) Sedimentation**

Sol. **(B) Filtration**
Tea leaves are insoluble and can be removed using a strainer/filter, which is filtration.

(iii) A mixture contains iron nails and sawdust. The best method to separate iron nails is:
(A) Filtration **(B) Sedimentation**
(C) Magnetic separation **(D) Evaporation**

Sol. **(C) Magnetic separation**
Iron nails are magnetic, so a magnet attracts them and separates them from sawdust.

(iv) What is decantation? Explain with a suitable example in 4–5 lines.

Sol. Decantation is the process of pouring out the clear upper liquid after the insoluble solid has settled at the bottom. First, sedimentation occurs (solid settles). Then the clear liquid is carefully poured into another container without disturbing the solid.
Example: After mud settles in muddy water, the clear water is poured out.

(v) A student filters muddy water using a cloth but the water is still slightly muddy.
(i) Why did this happen?
(ii) What should the student use to get clearer water?

Sol. (i) Cloth has larger pores, so very fine mud particles pass through it.
(ii) The student should use filter paper (or a finer filter) because it has smaller pores and stops fine particles better.

Practice Questions - 5

(i) A student collects muddy water from a field and wants clearer water quickly without using any filter. He keeps the water in a bucket for some time and notices the mud collects at the bottom. Which pair of processes is involved if he now gently pours the clear water into another bucket?
(A) Filtration and evaporation
(B) Sedimentation and decantation
(C) Magnetic separation and filtration
(D) Condensation and decantation

(ii) In a school lab, a mixture of sand and water is passed through a funnel fitted with filter paper. After some time, clear water collects in the beaker and sand remains on the paper. The sand left on the filter paper is called:
(A) Filtrate (B) Residue
(C) Sediment (D) Solution

(iii) A student uses a cloth to filter muddy water but still finds the water slightly muddy. The best reason is:
(A) Cloth absorbs water completely
(B) Mud is soluble in water
(C) Cloth has larger pores so fine particles pass through
(D) Filtration works only for magnetic substances

(iv) A mixture contains iron pins, small stones and sawdust. The student wants to separate only the iron pins first, in the easiest way. Which method should be used?
(A) Filtration (B) Magnetic separation
(C) Sedimentation (D) Decantation

(v) Which of the following mixtures is best separated by sedimentation?
(A) Salt + water (B) Sugar + water
(C) Sand + water (D) Oil + water

(vi) The clear liquid collected after filtration is called:
(A) Residue (B) Filtrate
(C) Sediment (D) Ore

(vii) Magnetic separation is useful when:
(A) Both substances are soluble in water
(B) One substance is magnetic and the other is non-magnetic
(C) Both substances are liquids
(D) The mixture contains only gases

(viii) Explain sedimentation in 3–4 lines and write one daily life example where you see it.

(ix) Write the steps to separate a mixture of sand and water using filtration (4–5 points). Also name the residue and filtrate.

(x) You are given a mixture of iron filings and sand.
(i) Which method will you use to separate them?
(ii) Write the procedure in 4–5 lines.

6. SOLUBILITY AND SOLUTIONS

6.1 Key Definitions: Solute, Solvent and Solution

(a) Solute: A solute is the substance that dissolves in a liquid.

Examples: Sugar and salt dissolve in water, so they are solutes.

(b) Solvent: A solvent is the liquid that dissolves the solute.

In most common examples, water is the solvent because many substances dissolve in it

(c) Solution: A solution is a uniform mixture formed when a solute dissolves completely in a solvent. In a solution, the solute particles spread evenly throughout the solvent.

Example: Sugar dissolved in water forms a sugar solution.

Real-Life Example: ORS Solution

ORS (Oral Rehydration Solution) is prepared by dissolving sugar and salt in water.

Composition:

- Solute: Sugar and salt
- Solvent: Water
- Solution: ORS

This example shows how solute and solvent combine to form a useful solution.

6.2 Types of Solutions

Solutions are classified based on how much solute can dissolve in the solvent.

(a) Unsaturated Solution: An unsaturated solution is a solution in which more solute can still dissolve at the same temperature.

How to identify: If more solute is added and it dissolves completely, the solution is unsaturated.

Characteristics:

- Contains less solute than the maximum possible amount
- Remains clear and uniform
- Has the capacity to dissolve more solute

Example: Water with a small amount of sugar that can still dissolve more sugar.

(b) Saturated Solution: A saturated solution is a solution in which no more solute can dissolve at the same temperature.

How to identify: If more solute is added and it does not dissolve and settles at the bottom, the solution is saturated.

Characteristics:

- Contains the maximum amount of solute
- Extra solute remains undissolved
- The solvent has reached its full dissolving capacity

Example: A strong sugar solution in which extra sugar does not dissolve.

Examples

(i) When sugar is mixed in water and it "disappears" after stirring, sugar acts as:
(A) Solvent **(B) Solute**
(C) Residue **(D) Sediment**

Sol. **(B) Solute**
Sugar dissolves in water. The substance that dissolves is the solute.

(ii) In a salt solution (salt + water), the correct pair is:
(A) Solute: water, Solvent: salt
(B) Solute: salt, Solvent: water
(C) Solute: salt, Solvent: salt
(D) Solute: water, Solvent: water

Sol. **(B) Solute: salt, Solvent: water**
Salt is the substance that dissolves → solute; water dissolves it → solvent.

(iii) Which pair of liquids is immiscible (does not mix completely and forms layers)?
(A) Vinegar + water **(B) Honey + water**
(C) Oil + water **(D) Lemon juice + water**

Sol. **(C) Oil + water**
Oil does not mix with water and forms a separate layer, so they are immiscible.

(iv) Define solute and solvent with one example of each (4–5 lines).

Sol. **Solute**: The substance that dissolves in a liquid.
Example: sugar in water.
Solvent: The liquid that dissolves the solute.
Example: water dissolving sugar.
When solute dissolves in solvent, it forms a solution.

(v) "Oxygen dissolved in water is important for aquatic life."
Which type of solution is this?

Sol. This is a gas in liquid solution.
Solute: oxygen gas
Solvent: water
Because oxygen mixes in water by dissolving, water contains dissolved oxygen which helps fish and other aquatic organisms to breathe.

Practice Questions - 6

(i) A student adds a spoon of salt to a glass of water and stirs. After some time, the mixture looks uniform and clear. Which option correctly identifies solute and solvent in this case?
(A) Solute: water, Solvent: salt (B) Solute: salt, Solvent: water
(C) Solute: water, Solvent: water (D) Solute: salt, Solvent: salt

(ii) Riya mixes a little sand in water and stirs well. Even after stirring, sand remains visible and settles down after some time. Which conclusion is correct?
(A) Sand is soluble in water (B) Sand is the solvent
(C) Sand is insoluble in water (D) Water becomes solute

(iii) Two liquids A and B are mixed in a test tube. After shaking and keeping it still, two separate layers are seen. What does this observation indicate?
(A) The liquids are miscible (B) The liquids are immiscible
(C) One liquid is a solid (D) A gas has dissolved

(iv) A fish survives in a pond because a gas is dissolved in water. Which option correctly describes the type of solution and the dissolved substance?
(A) Solid in liquid; salt (B) Gas in liquid; oxygen
(C) Liquid in liquid; oil (D) Gas in solid; oxygen

(v) Which of the following is an example of a liquid in liquid solution (miscible liquids)?
(A) Sugar in water (B) Oxygen in water
(C) Vinegar in water (D) Sand in water

(vi) In a solution, the component present in larger amount is generally called the:
(A) Solute (B) Solvent
(C) Residue (D) Sediment

(vii) Which statement is correct?
(A) Miscible liquids always form two layers
(B) Immiscible liquids mix completely
(C) Miscible liquids form a single uniform layer
(D) Immiscible liquids are always solids

(viii) Define solution in your own words and give one example (4–5 lines).

(ix) Write two examples each of:
(i) Soluble substances in water
(ii) Insoluble substances in water

(x) Oil and water are mixed in a bottle. After some time, two layers appear.
(i) Name this type of liquids (miscible/immiscible).
(ii) Which layer will be on top in general?

7. SOLUBILITY OF GASES AND DENSITY

7.1 Gases Dissolved in Liquids

What does "gas dissolved in liquid" mean?
When a gas mixes uniformly in a liquid and does not form a separate layer, the gas is said to be dissolved in the liquid.

Different gases dissolve in water in different amounts.

Important Example: Oxygen in Water

Oxygen dissolves in water and is essential for the survival of aquatic animals such as fish. Aquatic plants also use dissolved gases for their life processes. If oxygen levels decrease, aquatic organisms may not survive.

Daily-Life Examples

- Cold soft drinks contain dissolved carbon dioxide gas. When the bottle is opened, gas escapes as bubbles.
- In aquariums, air pumps are used to increase the amount of dissolved oxygen in water

7.2 Factors Affecting Dissolution of Gases in Liquids

(a) Pressure: Higher pressure increases the amount of gas dissolved in liquid.

Example: Soft drinks are packed under high pressure.

(b) Temperature: Lower temperature helps more gas remain dissolved.

Example: Cold drinks remain fizzy longer.

(c) Shaking or Stirring: Shaking helps gas escape faster from the liquid.

Example: When you shake a cold drink bottle (like soda) and open it, gas escapes quickly and bubbles come out fast.

7.3 Density , Sinking and Floating

Recall: Mass and Volume

- **Mass**: Amount of matter present in an object
- **Volume**: Space occupied by an object

Density: Density is the amount of mass present in a given volume of a substance. It tells us how tightly matter is packed.

Formula:

Density = Mass ÷ Volume

Units:

- g/cm^3 (commonly used in school)
- kg/m^3 (SI unit)

Important Points About Density:

- For the same volume, more mass means higher density.
- For the same mass, more volume means lower density.
- Density helps compare different materials fairly.

7.4 Sinking and Floating Based on Density

Whether an object sinks or floats depends on its density compared to water.

(a) Floating Condition : If the density of an object is less than the density of water, it floats.

Density (object) < Density (water) → Object floats

Example: Wood floats on water.

(b) Sinking Condition: If the density of an object is greater than the density of water, it sinks.

Density (object) > Density (water) → Object sinks

Example: Stone sinks in water.

(c) Suspended Condition: If the density of an object is nearly equal to the density of water, it may remain suspended in the middle.

Density (object) ≈ Density (water) → Object remains suspended

Classroom-Friendly Examples

- Stone in water → sinks (higher density)
- Wood in water → floats (lower density)
- Oil in water → floats (lower density than water)

Examples

(i) When Magnesium ribbon is burnt in the presence of Oxygen, it forms a white powder called M Which gas dissolved in water is most important for the survival of fish and other aquatic animals?

(A) Nitrogen **(B) Oxygen**
(C) Carbon dioxide **(D) Hydrogen**

Sol. **(B) Oxygen**
Aquatic animals depend on dissolved oxygen present in water.

(ii) Two objects A and B have the same volume. Object A has greater mass than object B. Which statement is correct?

(A) A is less dense than B
(B) A is more dense than B
(C) A and B have the same density
(D) Density cannot be compared without water

Sol. **(B) A is more dense than B**
For the same volume, greater mass means higher density.

(iii) An object is placed in water. It floats. Which statement is most correct?

(A) Object is denser than water
(B) Object has no mass
(C) Object is less dense than water
(D) Water is denser than all substances

Sol. **(C) Object is less dense than water**
Objects with density less than water generally float.

(iv) A student says, "Big objects always sink." Explain why this statement is wrong using the idea of density.

Sol. Sinking or floating depends mainly on density, not only on size.
Density means mass per unit volume.
A big object can float if its average density is less than water
Example: A ship is very large, but it floats because its average density (including air spaces) is low.

(v) What does it mean when an object remains suspended in water (neither sinks nor floats)? Explain using density

Sol. If an object stays suspended, it means it does not move up or down in water.
This happens when the object's density is nearly equal to the density of water.
So it neither becomes clearly lighter (float) nor clearly heavier (sink).

Practice Questions: - 7

(i) What does "gas dissolved in liquid" mean?
(A) Gas forms a separate layer in liquid
(B) Gas mixes uniformly in liquid
(C) Gas freezes in liquid
(D) Gas changes into solid

(ii) The gas essential for aquatic animals like fish is —
(A) Nitrogen (B) Hydrogen
(C) Oxygen (D) Helium

(iii) Why are soft drinks packed under high pressure?
(A) To change their colour
(B) To increase dissolved gas in them
(C) To decrease temperature
(D) To remove sugar

(iv) More gas remains dissolved in a liquid at —
(A) High temperature (B) Low temperature
(C) Continuous heating (D) Shaking

(v) Density is defined as —
(A) Mass × Volume (B) Mass ÷ Volume
(C) Volume ÷ Mass (D) Mass + Volume

(vi) An object will float on water if its density is —
(A) Equal to water (B) Greater than water
(C) Less than water (D) Zero

(vii) An object with density nearly equal to water will —
(A) Float on surface
(B) Sink to bottom
(C) Remain suspended in middle
(D) Dissolve in water

(viii) Why is dissolved oxygen in water important for aquatic animals?"
Also mention one way we can increase oxygen in water in a fish tank.

(ix) Student drops three objects X, Y and Z in water:
X floats on the surface
Y sinks to the bottom
Z stays in the middle
Explain what you can conclude about the density of X, Y and Z compared to water (write in points).

(x) A student has two objects A and B.
Mass of A = 200 g, Volume of A = 100 cm^3
Mass of B = 150 g, Volume of B = 100 cm^3
(i) Which object is denser?
(ii) Which one is more likely to sink in water? (Write reason in one line.)

Practice Questions - Answer Key

(1)	**(i)** (C)	**(ii)** (C)	**(iii)** (B)	**(iv)** (C)	**(v)** (C)	**(vi)** (A)	
(2)	**(i)** (B)	**(ii)** (C)	**(iii)** (B)	**(iv)** (B)	**(v)** (A)	**(vi)** (C)	
(3)	**(i)** (C)	**(ii)** (B)	**(iii)** (C)	**(iv)** (C)	**(v)** (B)	**(vi)** (C)	**(vii)** (C)
(4)	**(i)** (B)	**(ii)** (C)	**(iii)** (B)	**(iv)** (C)	**(v)** (A)	**(vi)** (B)	**(vii)** (C)
(5)	**(i)** (B)	**(ii)** (B)	**(iii)** (C)	**(iv)** (B)	**(v)** (C)	**(vi)** (B)	**(vii)** (B)
(6)	**(i)** (B)	**(ii)** (C)	**(iii)** (B)	**(iv)** (B)	**(v)** (C)	**(vi)** (B)	**(vii)** (C)
(7)	**(i)** (B)	**(ii)** (C)	**(iii)** (B)	**(iv)** (B)	**(v)** (B)	**(vi)** (C)	**(vii)** (C)